Archiving and Compression of Files

Archiving Files

Archiving is to combine a group of files organized in a tree structure into one file.

This enables easier handling, such as for backup or transfer purposes

This is a separate procedure from compression

The tool used for archiving files is “tar”

Archiving Files (tar Command)

```
$tar <options> <destination archive file> <dirs/files to arch
ive>
```

This command is used to archive a group of files/dirs into a single tar file

It is also used to un-archive a tar file into its original dir tree structure.

To archive (tar) a group of files and dirs,

$ tar cvf <archiveFile.tar> <a set of files and dirs>

‘c’ for create

‘v’ for verbose

‘f’ for file

Examples:

$ tar cvf my-docs.tar ~/Documents/pdfs/

$ tar cvf selected-files.tar ~/files-1.txt ~/Documents/file-2.txt ~/*.pdf

Working with a tar file (tar Command)

- Now we have the tar file, and we can do the following with it,
  - To un-archive (un-tar) a file into all of its original components
    - $ tar xvf <archiveFile.tar>
  - To extract some of the files/dirs from a tar file
    - $ xcf <archiveFile.tar> <files/dir to untar>
  - To show the contents of a tar file
    - $ tar tvf <archiveFile.tar>
  - Examples:
    - `$tar cvf tar-file.tar *.pdf *.txt`
    - `$tar tvf tar-file.tar`
    - `$tar xvf tar-file.tar *.pdf`
    - `$tar xvf tar-file.tar`

Using tar inside a find Command

- Let us say, we need to **archive all pdf files** in our home directory.
- We will need the find command to search for those files
- The outcome of the find command is then passed to the tar command
- This can be achieved by using a **pipe**,
  - `$find ~ -name '*.pdf' | tar cvf file.tar`
- Another way to achieve the same target is to use the tar command as the user defined command to be executed inside the find command
  - `$find ~ -name '*.pdf' -exec tar cvf file.tar '{}' '+'`

Compressing Files

- Compressing files is the procedure of reducing the size of the file by a tool that removes any redundant data in the file.

- There are multiple formats for compressed files,
  - **'.gz'** → use the tools gzip and gunzip
    - $ gzip <file> file is compressed into file.gz
    - $ gunzip <file.gz> file.gz is flattened into file
  - Better compression '.bz2' → use the tool bzip2 and bunzip2
    - $bzip2 <file>
    - bunzip2 <file.bz2>
  - Even better compressino 'lzma' → use the tools lzma and unlzma
    - $lzma <file>
    - $unlzma <file.lzma>
- Note that these tools,
  - They deal with a single file and not a group of files, to be able to compress a group of file in a directory tree, you need to archive them first.
  - They replace the file with its compressed/flattened version. The original file is deleted.

Accessing Compressing Text Files

- If a text file is compressed to a `.gz` format, we can still access it without uncompressing it using a set of tools (**Z-tools**)

  `$gzip my-file.txt`
- To view the file contents, we use the `zcat` command (similar to cat command for uncompressed text files)
  - $ **zcat** my-file.gz
- If the file is **long**, and we need to **view** it **page by page**, we can use the commands `zmore` and `zless` (similar to more and less commands for uncompressed text commands)
- We can also search inside the compressed text doc via the zgrep and zegrep commands (similar to grep & egrep)

More on compressing files

- Note that compression preserves the permissions and timestamp of the files, After uncompressing the compressed file, we end up the same permissions and timestamps of the original file.
- It is not useful to try to compress an already compressed file, it will probably increase its size due to
  - No reduction in file size since reduction already happened the first time.
  - Added more overhead data (meta-data) at the second compression time)

Mixing Archive and Compression

- Archiving means combining multiple files organized in a tree structure into a single file.
- Compression is to reduce the size of a single file.
- So if we want to perform both, we can do this in two steps,
  - $tar cvf pdf-files.tar *.pdf (this will generate the pdf-files.tar)
  - $gzip pdf-files.tar (this will generate the pdf-file.tar.gz)
- We can replace the gzip with any other compression tool (depending on the desired compression format)
- We can perform both tasks (arch & comp) in a single step

Using tar for Archive + Compress

```
$ tar <options> <compressed file><files or folders>
$ tar <options> <compressed file>
```

- With the right set of options to the tar command, we can perform both archiving and compression (to the desired compression format)
  - Add the option ‘z’ to perform gzip or gunzip

    - Add the option ‘j’ to perform bzip2 or bunzip2
    - Add the option ‘—lzma’ to perform lzma or unlzma
- Examples

```
$tar cvzf my-file.tar.gz ~/Documents/*.doc
$tar xvzf my-file.tar.gz
$tar xvjf file.tar.bz2
$tar cvf –lzma my-file.tar.lzma ~/my-project
```

Other Archive + Compression Tools (the rar Command)

- The `rar` tool can be used to **archive + compress** or **extract** + **deflate** files to/from the **.rar** format archive file
- The tool need to be installed first
    - `$sudo apt-get install rar`
- The rar tool is very powerful tool, we will only cover the basics of it
- To add a file (group of files) to a rar archive
    - `$rar a my-archive.rar ~/projects/*.pdf`
    - This will add the pdf files to the my-archive.rar archive.
        - If the archive does not exist, it will be created.
        - It archive exists, it will be appended with these files.
    - To lock the archive to stop more file additions
        - `$ rar k my-archive.rar`